Linux determines the accessibility of files depending on two important factors that define it:-

- File ownership
- File Permissions

A complete understanding of File Ownership and File Permission is essential for any Linux Programmer who wants to excel and do complex coding in this operating system. We'll ensure whether you are familiar or not so familiar with Ubuntu and Linux you get the basics of this chapter right.

File Ownership in Ubuntu

Note: File Ownership is just a term that defines the concept of file ownership and it is applicable to directories as well. Every File or Directory in Ubuntu and other Linux systems have 3 types of owners, namely User, Group and Others.

1. User

User is the first and the principle owner of a file. This is the entity which creates the file and thus gets the ownership of it. However, ownership can also be changed via User Permissions. We'll look into this a bit later.

2. Group

Every user is a part of numerous groups depending on the user permissions of numerous files and directories. Grouping user with similar permissions makes it easier to manage user in the multi-user environment. This concept can be further explained by an example which is given below:-

Consider that you are an owner of a company which has three teams, namely the Development team, sales team and Finance team, with all of them accessing the same system from their respective locations. Now what if you club each of the members of the teams in a single group and give all of them permission to access particular files, folders and directories depending on their level and their team? This would save a lot of time that might have been used up in granting permission to each and every user based on his level and team.

This way even if you are the principle user, you'll still be a part of all the groups and the only person with administrative functions until you change the User Permissions and File Access yourself.

Note: You can try and run the command groups to determine which all user groups you are added to.